MIL-R-22763(WP)
AMENDMENT-2
1 APRIL 1965
SUPERSEDING
AMENDMENT-1
22 DECEMBER 1961

MILITARY SPECIFICATION

RECEIVER-TRANSMITTER, RADIO RT-547(*)/ASQ-19
AND
DECODER, PULSE KY-312(*)/ASQ-19

This amendment forms a part of Military Specification MIL-R-22763(WEP) dated 20 February 1961 and has been approved by the Bureau of Naval Weapons, Department of the Navy.

*Page 1, paragraph 1.2: Add

"Indicator, course ID-387/ARN"

Page 1, paragraph 2.1, after "invitation for bids": Add:

"except the dated document indicated below,"

Page 1, paragraph 2.1, under SPECIFICATIONS Military: Add:

"MIL-I-6181B (29 May 1953)	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment (See 2.2(2))"

Page 2, paragraph 2.2: Add a new subparagraph

"(2) Specification MIL-I-6181B is available only from Bureau of Naval Weapons, Washington 25, D. C. Attention RREN-63."

*Page 2, paragraph 3.2.5.1: 1st sentence: delete "Type DA" and substitute "Type D". Delete last two sentences and substitute:

"External connectors shall be Amphenol types 126-10572, 126-10753 and Industrial Products type 81450 or approved equivalents. The wires and terminals of the type 126-10753 connector shall be potted with a sealing compound which is capable of withstanding temperatures to 125°C."

Page 2, Add the following paragraphs.

"3.2.8 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Military Specification MIL-I-6181B with the following clarifications and additions.

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*3.2.8.1 Conducted Interference (Power Lead) - Power lead conducted interference shall be controlled within the limits of Specification MIL-I-6181B, except as follows:

Frequency Range	Level
0.15 to 0.25 mc	3000 uv/kc
0.25 to 20 mc	1000 uv/kc and 1000 uv-cw

*3.2.8.2 Radiated Interference - Radiated interference shall be controlled within the limits of Specification MIL-I-6181B, except at the frequencies listed below. Radiated interference shall not exceed the indicated levels:

Frequency Range	Level
0.15 to 0.25 mc	30 uv/kc
20 to 50 mc	1 uv/kc
1/3 transmit frequency	200 uv-cw
1/2 transmit frequency	200 uv-cw
2/3 transmit frequency	200 uv-cw

3.2.8.3 Antenna Conducted Interference

(1) Antenna conducted interference in the frequency range 0.15 to 960 mc shall be controlled within the limits of Specification MIL-I-6181B except at a frequency corresponding to one-third the transmit frequency. The level shall not exceed 500 uv.

(2) The antenna conducted cw power from the equipment, between transmitted pulses or in the receive mode of operation in the frequency range of 1015 to 1045 mc shall not exceed a level 100 db below the peak power of transmitted pulse.

(3) At frequencies between 1250 and 10,000 mc, the antenna conducted output of the transmitter under transmit conditions shall not exceed a level of -30 dbw except that, at frequencies corresponding to the third and fifth harmonics of the transmitter frequency, the antenna conducted interference shall not exceed a level of -10 dbw.

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3.2.8.4 Receiver Front End Rejection - The equipment shall be capable of rejecting the following types of signal under the conditions specified.

3.2.8.4.1 CW Frequency - With the equipment tracking on a signal of -105 dbw, the specified performance shall not be affected by the following types and levels of interference at any frequency on or within 3 mc of the desired frequency.

Type	Level
(1) 6000 randomly spaced single pulses per second, each 3.5 microseconds wide.	-105 dbw
(2) CW	-120 dbw

*3.2.8.4.2 OFF Frequency - With the equipment tracking on a signal of -105 dbw, the specified performance shall not be affected when cw signals of the following frequencies and amplitudes are applied to the antenna connector through an appropriate filter:

Frequency	Amplitude
0.1 to 950 mc	-60 dbw
63 mc (IF)	-40 dbw
Image frequency	-60 dbw
1250 to 2000 mc	-60 dbw
2000 to 3000 mc	-90 dbw
3000 to 4000 mc	-70 dbw
4000 to 7000 mc	-80 dbw
7000 to 10,000 mc	-60 dbw

*Page 2, paragraph 3.2.12, after line 3: Delete and substitute

"Cooling Air (1) RT-547/ASQ-19 19.98 lbs/hr of ambient room air.
(2) KY-312/ASQ-19 37.08 lbs/hr of ambient room air.
B+ and Bias Voltages 430 ± 12V DC, 250 ± 7V DC, 130 ± 4V DC from -265 ± 5 VDC source through external series resistor of 4000 ± 10% ohms.
Transmitter Load 50 ohms resistive."

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*Page 2, paragraph 3.2.13.1, line 3: Add

"and Figure 2"

*Page 3: Delete Figure 1 and add Figures 1 and 2 of this amendment.

*Page 4, paragraph 3.3.1.2: Delete "52" and substitute

"50"

*Page 6, paragraph 3.3.1.10: Delete last sentence

*Page 6, paragraph 3.3.1.11.2: Add

"The 8 db difference shall be measured using test signals, each having an equal modulation percentage but not to exceed a total of 40% modulation."

*Page 7, paragraph 3.3.1.15, after standard conditions: Add to beginning of first sentence

"(Upon initial adjustment or when recalibration is necessary.)"

*Page 7, paragraph 3.3.1.17.5: Delete and substitute

"3.3.1.17.5 Interrogation Pulse Repetition Rate - When the equipment is operating in the T/R mode, the pulse repetition rate shall not exceed 150 pulse pairs per second (pp/s) in the search condition, and shall not exceed 30 pulse pairs per second (pp/s) in the track condition. The pulse repetition rate shall vary or jitter enough to preclude synchronization of transmitted interrogations between any of 100 equipments on the same channel for more than one second in one hour. A search repetition rate of 135 pp/s shall be avoided to preclude synchronization with the fine bearing information."

*Page 8, paragraphs 3.3.1.20.2 and 3.3.1.21.3.3: Delete

*Page 9, paragraph 3.3.1.22.1: Delete and substitute

"3.3.1.22.1 Bearing Flag Alarm - The bearing flag alarm circuit shall provide a terminal which furnishes +28V DC to the ID-387/APN flag alarm when the bearing circuits are properly tracking a TACAN signal or are in memory, and which is open circuited when the bearing circuits are not properly tracking. This terminal shall also be open circuited in the STBY mode of operation."

*Page 9, paragraph 3.3.1.22.2: Delete and substitute

"3.3.1.22.2 Distance Flag Alarm - The distance flag alarm circuit shall provide a terminal which is grounded when the distance circuits are properly tracking a TACAN signal or are in memory, and which is ungrounded when the distance circuits are not properly tracking. A terminal shall also be provided to furnish +28V DC to the distance flag alarm in the ID-663/U."

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*Page 9, paragraph 3.3.1.23.1: Delete and substitute

"3.3.1.23.1 Blanking Pulse Trigger Ouput - This equipment shall generate a single blanking pulse trigger, with characteristics specified below, for each interrogation pulse pair transmitted. This trigger shall normally be used in the AN/ASQ-19 system to initiate suppression of the IFF receiver.

(1) Polarity: Positive
(2) Amplitude: 15 to 70 volts
(3) Load Impedance: 2200 ohms in parallel with 500 picofarads at the blanking pulse trigger output terminal.
(4) Rise Time: 0.7 microsecond maximum from 3 volt to 10 volt level.
(5) Duration: 1.5 to 20 microseconds at 10 volt level; duration at the 3 volt level shall not exceed 30 microseconds.
(6) Synchronization: The 10 volt level on the leading edge shall occur at least 0.5 but not more than 5.0 microseconds before the point at which the first pulse of the transmitted interrogation pulse pair reaches 10% of its maximum voltage amplitude.
(7) Spurious Output: Any spurious output at the blanking pulse trigger terminal shall not exceed ± 1.0 volt in amplitude with the specified load impedance. Spurious output shall be defined as any output outside of the maximum dimensions of the blanking pulse trigger envelope permitted above, with an additional 30 microseconds allowed for the decay from the 3 volt to the 1 volt level.
(8) Operating Mode: Blanking pulse triggers shall be generated only when the equipment is in the T/R mode of operation."

*Page 10, paragraph 3.3.1.24: Delete paragraph and substitute

"3.3.1.24 Power Requirements - The equipment shall give specified performance with the power sources listed in Specification MIL-I-18309. The power requirements shall not exceed the specified amounts.

(1) A-C Power, Phase 2 (to neutral)

26-Volts A-C	1.5 amperes
115-Volts A-C	2.2 amperes

(2) A-C Power, Phases 1 and 3 (to neutral)

115-Volts A-C	0.120 ampere

(3) D-C Power 28-Volt 8.0 amperes, intermittent (channeling)

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(4) D-C Power (from AN-2349/ASQ-19)

(a)	+430 V	
	Operating Voltage Limits	378 to 495 volts
	Current	0.006 ampere at 430 volts
(b)	+250 V	
	Operating Voltage Limits	221 to 274 volts
	Current	0.034 ampere at 250 volts
(c)	+140 V	
	Operating Voltage Limits	137 to 143 volts
	Current	.220 ampere at 140 volts
(d)	+130 V	
	Operating Voltage Limits	117 to 140 volts
	Current	0.094 ampere at 130 volts
(e)	-180 V	
	Operating Voltage	-238 to -300 volts source through external series resistor of 4000 ±10% ohms
	Current	0.024 ampere when source is -265 volts"

*Page 11, paragraph 3.4.1.1: Add

"(4) Provide blanking pulse triggers in accordance with 3.3.1.23.1."

*Page 11, paragraph 3.4.1.2: Delete and substitute

"3.4.1.2 Form Factor - The radio receiver-transmitter shall be contained in a single unit capable of being directly mounted to a shockmounted equipment shelf which in turn is mounted in an aircraft. The over-all dimensions, excluding protrusions shall not exceed

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7-17/32 by 7-37/64 by 22-37/64 inches. The front panel shall extend an additional 1-1/32 inches below the base. Protrusions of 17/64 inches shall be allowed on each side in addition to the above dimensions."

*Page 11, paragraph 3.4.1.4, under "Receptacle Type": Delete "126-10753" and substitute

"126-10572."

*Page 11: Add

"3.4.1.6 Pressurization - The case containing this equipment requires a supply of dry pressurized air. At an internal pressure of 15 psi gauge, the case construction shall be such that the leakage is no greater than 0.5 psi in 2 hours. The equipment performance shall be in accordance with this specification when the internal pressure is maintained at 20 ± 5 psia."

*Page 12, paragraphs 3.4.2.1(10) and 3.4.2.1(11): Delete

*Page 12, paragraph 3.4.2.2: Add

"Protrusions of 5/64 inches shall be allowed on each side in addition to the above dimensions."

*Page 14, paragraph 3.4.2.5: Delete "Amphenol 126-10572" and substitute

"Amphenol 126-10753."

*Page 14, paragraph 4.1: Delete and substitute

"4.1 General - Equipments shall be subjected to the test requirements of MIL-I-18309 to determine compliance with all applicable requirements."

An asterisk preceding a page number denotes a change to be made to the specification which did not appear in Amendment-1.

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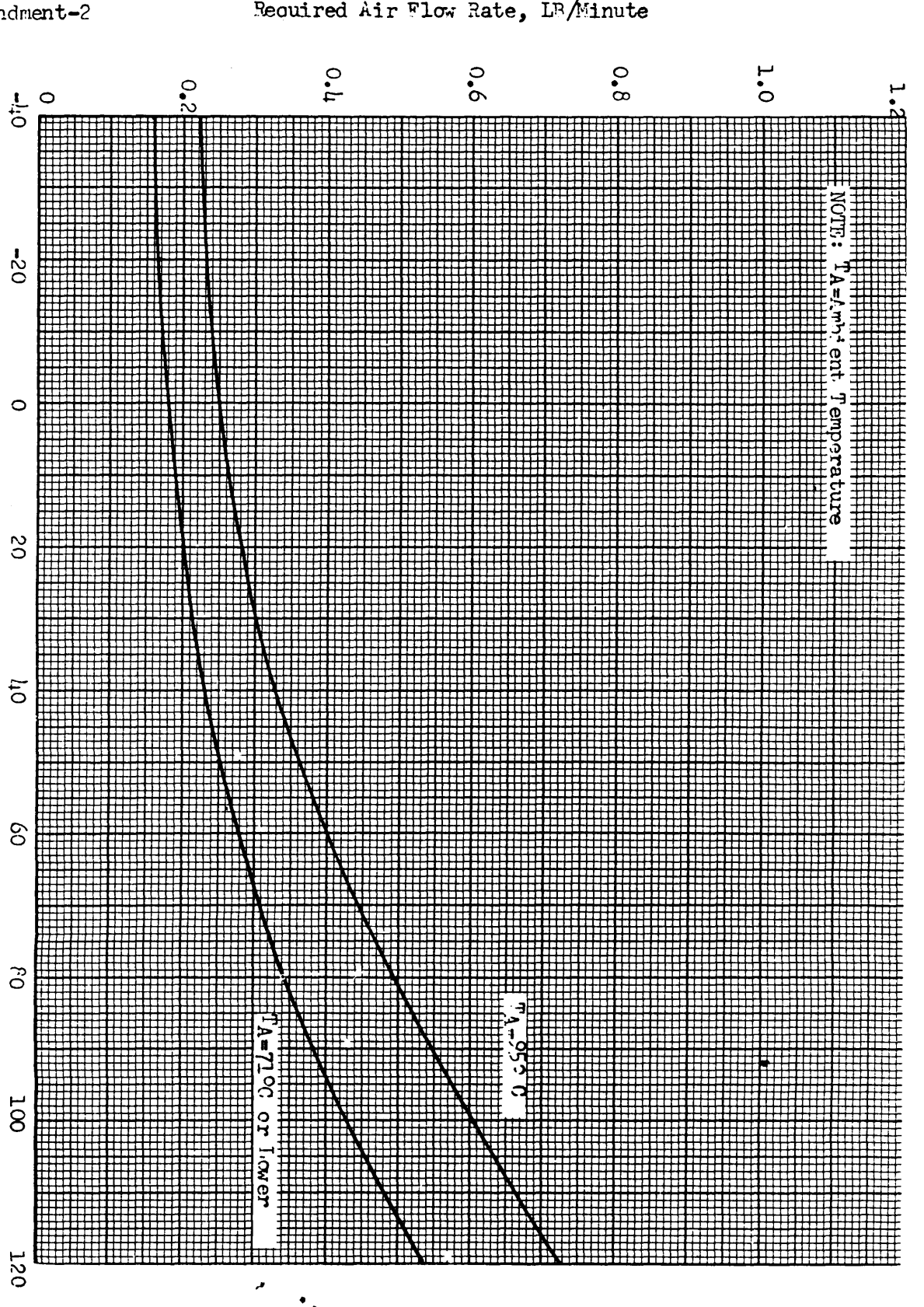


Figure 1. RT-547/ASQ-19 Receiver-Transmitter Minimum Cooling Air Requirements

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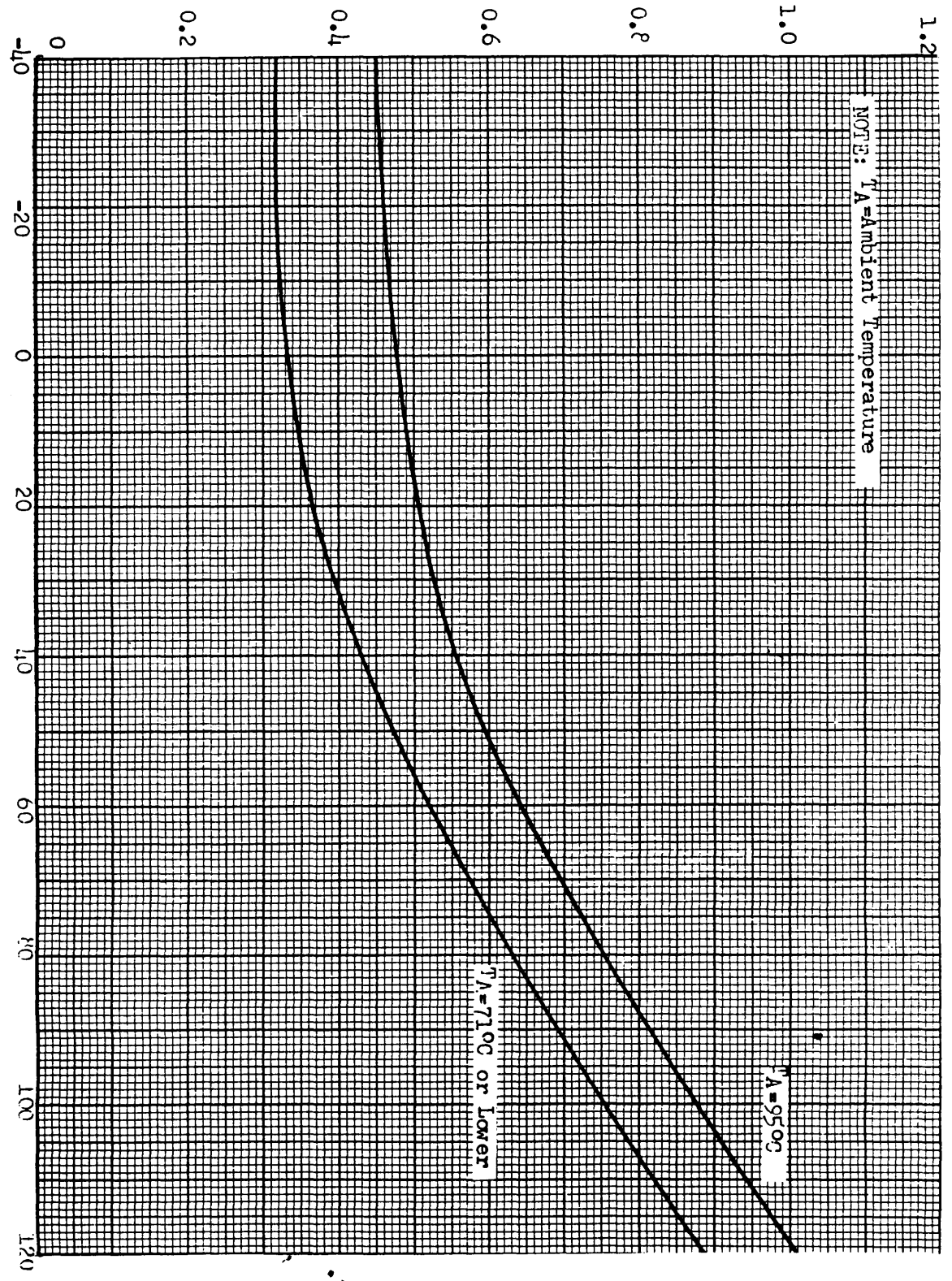


FIGURE 2. KY-312/ASQ-19 PULSE DECODER MINIMUM COOLING AIR REQUIREMENTS

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